HW #01

2013

P1 standard textbook material → read for yourself.

$$\hat{S}_{\theta,\phi} \Rightarrow \begin{vmatrix} \frac{1}{2}\cos\theta - \lambda & \frac{1}{2}\sin\theta\, e^{-i\phi} \\ \frac{1}{2}\sin\theta\, e^{i\phi} & -\frac{\cos\theta}{2} - \lambda \end{vmatrix} = 0$$

P2

$$-\frac{\cos^2\theta}{4} + \lambda^2 - \frac{\sin^2\theta}{4} = 0 \qquad \therefore \lambda = \pm\frac{1}{2}$$

$\lambda = \frac{1}{2}$ $\quad \therefore$

$$X(\cos\theta - 1) + Y\sin\theta \cdot e^{-i\phi} = 0$$
$$X\sin\theta\, e^{i\phi} - Y(\cos\theta + 1) = 0$$

$$\cos\theta = 2\cos^2\frac{\theta}{2} - 1 \qquad \therefore \quad X e^{i\phi}\sin\frac{\theta}{2} = Y\cos\frac{\theta}{2}$$

$$\sin\theta = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2} \qquad \Rightarrow \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\, e^{i\phi} \end{pmatrix}$$

$\lambda = -\frac{1}{2}$ $\quad X(\cos\theta + 1) + Y\sin\theta\, e^{-i\phi} = 0 \quad \therefore X\cos\frac{\theta}{2} + Y\sin\frac{\theta}{2}e^{-i\phi} = 0$

$$\Rightarrow \begin{pmatrix} \sin\frac{\theta}{2} \\ -\cos\frac{\theta}{2}\, e^{i\phi} \end{pmatrix}$$

$$S_{\theta,\phi} = \frac{1}{2}\begin{pmatrix} c & se^{-i\phi} \\ se^{i\phi} & -c \end{pmatrix} \qquad S^2_{\theta,\phi} = \frac{1}{4}\begin{pmatrix} c & se^{-i\phi} \\ se^{i\phi} & -c \end{pmatrix}\begin{pmatrix} c & se^{-i\phi} \\ se^{i\phi} & -c \end{pmatrix}$$

$$S^2_{\theta,\phi} = \frac{1}{4}\begin{pmatrix} c^2+s^2 & 0 \\ 0 & c^2+s^2 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

P3

$$\frac{d}{dt}\langle \hat{\underline{S}} \rangle = \frac{1}{i\hbar}\langle [\hat{\underline{S}}, \underbrace{\mathcal{H}}_{g\mu_B B \hat{S}_z}] \rangle$$

$$= \frac{g\mu_B B}{i\hbar}\langle [\hat{\underline{S}}, \hat{S}_z] \rangle$$

now $[S_z, S_z] = 0 \quad [S_x, S_z] = -i S_y \quad [S_y, S_z] = i S_x$

$$\frac{d}{dt}\langle \hat{\underline{S}} \rangle = \frac{g\mu_B B}{\hbar}\begin{pmatrix} -S_y \\ +S_x \\ 0 \end{pmatrix} \qquad \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ S_x & S_y & S_z \\ 0 & 0 & B \end{vmatrix} = \begin{pmatrix} S_y B \\ -S_x B \\ 0 \end{pmatrix}$$

$$= -\frac{g\mu_B}{\hbar}\langle \hat{\underline{S}} \rangle \times \underline{B}$$

(homework)

P4 Spin ½ + Orbital angular momentum ℓ

$\vec{J} = \vec{L} + \vec{S}$. $\vec{S}^2 = \frac{3}{4}\hbar^2$

Old states are $|\ell, m_\ell ; s, m_s\rangle \Rightarrow |\psi_{m_\ell m_s}\rangle$

New states $|j, m ; \ell, s\rangle \Rightarrow |\phi_{jm}\rangle$, $m = m_\ell + m_s$

Two possibilities: $\begin{cases} j = \ell + \frac{1}{2} & \text{ladder contains } 2j+1 = \underline{2\ell+2} \text{ states} \\ j = \ell - \frac{1}{2} & \text{ladder contains } 2j+1 = 2\ell - 1 + 1 = \underline{2\ell} \end{cases}$

$\Rightarrow$ total # states $2\ell + 2 + 2\ell = 2(2\ell+1)$ which is exactly the dimension of $|\psi\rangle$ space.

Start with $\underline{\underline{j = \ell + \frac{1}{2}}}$: $|\phi_{\ell+\frac{1}{2}, m}\rangle = \sum_{m_\ell + m_s = m} c_{m_\ell m_s} |\psi_{m_\ell, m_s}\rangle =$

$$= \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle$$

$m_\ell = m - \frac{1}{2}$, $m_s = \frac{1}{2}$ $\qquad$ $m_\ell = m + \frac{1}{2}$, $m_s = -\frac{1}{2}$

Normalization: $|\alpha_m|^2 + |\beta_m|^2 = 1$

Apply lowering operator $\hat{J}_- |\phi_{\ell+\frac{1}{2}, m}\rangle \overset{\text{by def}}{=} \hbar\sqrt{j(j+1) - m(m-1)}\, |\phi_{\ell+\frac{1}{2}, m-1}\rangle$

(here $j = \ell + \frac{1}{2}$)

$$= \hbar\sqrt{(j+m)(j-m+1)}\, |\phi_{\ell+\frac{1}{2}, m-1}\rangle = \hbar\sqrt{(\ell+m+\tfrac{1}{2})(\ell-m+\tfrac{3}{2})} \cdot$$

$(j = \ell + \frac{1}{2})$

$$\cdot \left\{ \alpha_{m-1} |\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \beta_{m-1} |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right\} . \quad (1)$$

On the other hand

$$\hat{J}_- |\phi_{\ell+\frac{1}{2}, m}\rangle = (\hat{L}_- + \hat{S}_-) \left\{ \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle \right\}$$

$$= \hbar\, \alpha_m \sqrt{\ell(\ell+1) - (m-\tfrac{1}{2})(m-\tfrac{3}{2})} \left(|\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \sqrt{\tfrac{1}{2}(1+\tfrac{1}{2}) - \tfrac{1}{2}(\tfrac{1}{2}-1)}\, |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right)$$

$$+ \hbar\, \beta_m \sqrt{\ell(\ell+1) - (m+\tfrac{1}{2})(m-\tfrac{1}{2})} \left(|\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle + \underbrace{\sqrt{\tfrac{1}{2}(\tfrac{1}{2}+1) - (-\tfrac{1}{2})(-\tfrac{1}{2}-1)}}_{\emptyset}\, \underbrace{|\psi_{m+\frac{1}{2}, -\frac{1}{2}-1}\rangle}_{\text{does not exist}} \right)$$

Use $l(l+1) - m(m-1) = (l+m)(l-m+1)$.

Thus $\hat{J}_- |\phi_{l+\frac{1}{2},m}\rangle = \hbar\alpha_m \sqrt{(l+m-\frac{1}{2})(l-m+\frac{3}{2})}\, |\psi_{m-\frac{3}{2},\frac{1}{2}}\rangle +$

$+ |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle) + \hbar\, \beta_m \sqrt{(l+m+\frac{1}{2})(l-m+\frac{1}{2})}\, |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle$. (2)

Right-hand sides of (1) and (2) are the same $\Rightarrow$ we can compare coefficients:

$$\alpha_{m-1}\sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{3}{2})} = \alpha_m \sqrt{(l+m-\tfrac{1}{2})(l-m+\tfrac{3}{2})} \qquad (3)$$

(The relation for β_m obtains from normalization: $|\beta_m|^2 = 1 - |\alpha_m|^2$.)

Observe that (3):

$$\frac{\alpha_m}{\sqrt{l+m+\frac{1}{2}}} = \frac{\alpha_{m-1}}{\sqrt{l+m-\frac{1}{2}}} = r \quad (r \text{ is constant independent of } m!)$$

Now, for a maximum possible $m = l+\frac{1}{2}$:

$$|\phi_{l+\frac{1}{2},l+\frac{1}{2}}\rangle = \alpha_{l+\frac{1}{2}} |\psi_{l,\frac{1}{2}}\rangle + \beta_{l+\frac{1}{2}} |\psi_{l+1,-\frac{1}{2}}\rangle$$

state with $m_\ell = l+1$, $m_s = -\frac{1}{2}$.

But $m_\ell \le l$ always

$\Rightarrow \boxed{\beta_{l+\frac{1}{2}} = 0}$

$\Rightarrow \alpha_{l+\frac{1}{2}} = 1$.

$$\Rightarrow r = \frac{\alpha_{l+\frac{1}{2}}}{\sqrt{l+l+\frac{1}{2}+\frac{1}{2}}} = \frac{1}{\sqrt{2l+1}}$$

$$\Rightarrow \alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$$

$$|\beta_m|^2 = 1 - \frac{l+m+\frac{1}{2}}{2l+1} = \frac{l-m+\frac{1}{2}}{2l+1} \Rightarrow \beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$$

Thus $|\phi_{l+\frac{1}{2},m}\rangle = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle + \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$

The states from the other ladder, with $\underline{\underline{j=l-\frac{1}{2}}}$;

must be orthogonal to $|\phi_{l+\frac{1}{2},m}\rangle$ at the same $\underline{m}$:

Thus

$$|\phi_{l-\frac{1}{2},m}\rangle = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle - \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

Clearly $\langle\phi_{l+\frac{1}{2},m}|\phi_{l-\frac{1}{2},m}\rangle = 0$ always.

Thus we found Clebsch-Gordan coefficients for $\vec{J} = \vec{L} + \vec{S}$, $S = \frac{1}{2}$.

They are $\alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$ and $\beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$.

<u>The limit $l=\frac{1}{2}$:</u> $\alpha_m = \sqrt{\frac{1+m}{2}}$ $\quad \beta_m = \sqrt{\frac{1-m}{2}}$

$$|\phi_{l=1,m}\rangle = \sqrt{\frac{1+m}{2}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle + \sqrt{\frac{1-m}{2}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

$$|\phi_{0,m}\rangle = \sqrt{\frac{1-m}{2}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle - \sqrt{\frac{1+m}{2}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle =$$

$$= (\text{but } m=0 \text{ the only option}) = \frac{1}{\sqrt{2}}\left(|-\tfrac{1}{2},\tfrac{1}{2}\rangle - |\tfrac{1}{2},-\tfrac{1}{2}\rangle\right)$$

Singlet!

<u>triplet</u>
$m=+1,0,-1$

$$\begin{cases} |\phi_{1,+1}\rangle = |\frac{1}{2},\frac{1}{2}\rangle \\ |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}}\left(|-\frac{1}{2},\frac{1}{2}\rangle + |\frac{1}{2},-\frac{1}{2}\rangle\right) \\ |\phi_{1,-1}\rangle = |-\frac{1}{2},-\frac{1}{2}\rangle \end{cases}$$

Let now $J_1 = l = 1$, $J_2 = s = \frac{1}{2}$.

$$\Rightarrow \quad \alpha_m = \sqrt{\frac{1+m+\frac{1}{2}}{2+1}} = \sqrt{\frac{3+2m}{6}}, \quad \beta_m = \sqrt{\frac{3-2m}{6}}$$

$$\underbrace{|\phi_{j=\frac{3}{2},m}\rangle}_{|j,m\rangle \text{ state}} = \alpha_m \underbrace{|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle}_{|j_1=1,m_1;j_2=\frac{1}{2},m_2\rangle \text{ states}} + \beta_m |\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

$j=\frac{3}{2}$

$$|j=\tfrac{3}{2},m\rangle = \alpha_m |m_1 = m-\tfrac{1}{2};\tfrac{1}{2}\rangle + \beta_m |m_1 = m+\tfrac{1}{2}; m_2 = -\tfrac{1}{2}\rangle$$

4 states

$$|\tfrac{3}{2},\tfrac{3}{2}\rangle = |\underset{m_1}{1};\underset{m_2}{\tfrac{1}{2}}\rangle = \left(\alpha_{\frac{3}{2}} = 1,\ \beta_{\frac{3}{2}} = 0\right) = |J_1^z = 1\rangle|J_2^z = \tfrac{1}{2}\rangle$$

$$|\tfrac{3}{2},\tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}\,|0;\tfrac{1}{2}\rangle + \sqrt{\tfrac{1}{3}}\,|1;-\tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}\,|J_1^z=0\rangle|J_2^z=\tfrac{1}{2}\rangle + \sqrt{\tfrac{1}{3}}\,|J_1^z=1\rangle|J_2^z=-\tfrac{1}{2}\rangle$$

$$|\tfrac{3}{2},-\tfrac{1}{2}\rangle = \sqrt{\tfrac{1}{3}}\,|-1;\tfrac{1}{2}\rangle + \sqrt{\tfrac{2}{3}}\,|0;-\tfrac{1}{2}\rangle = \sqrt{\tfrac{1}{3}}\,|J_1^z=-1\rangle|J_2^z=\tfrac{1}{2}\rangle + \sqrt{\tfrac{2}{3}}\,|J_1^z=0\rangle|J_2^z=-\tfrac{1}{2}\rangle$$

$$|\tfrac{3}{2},-\tfrac{3}{2}\rangle = |-1;-\tfrac{1}{2}\rangle$$

Next, $j=\frac{1}{2}$ states: $j = |J_1 - J_2| = |1-\frac{1}{2}| = \frac{1}{2}$:

$$|\phi_{j=\frac{1}{2},m}\rangle = \sqrt{\frac{\frac{3}{2}-m}{3}}\,|j_1=1, m_1; j_2=\tfrac{1}{2}, m_2=\tfrac{1}{2}\rangle$$

$\hookrightarrow m_1 + m_2 = m$

$$- \sqrt{\frac{\frac{3}{2}+m}{3}}\,|j_1=1, m_1; j=\tfrac{1}{2}, m_2=-\tfrac{1}{2}\rangle$$

$\hookrightarrow m+\frac{1}{2}$

2 states $\begin{cases} \left|\frac{1}{2};\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\ \left|J_1^z=0\right\rangle\left|J_2^z=\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\ \left|J_1^z=1\right\rangle\left|J_2^z=-\frac{1}{2}\right\rangle \\ \left|\frac{1}{2};-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\ \left|J_1^z=-1\right\rangle\left|J_2^z=\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\ \left|J_1^z=0\right\rangle\left|J_2^z=-\frac{1}{2}\right\rangle \end{cases}$

Thus total of $\left.\begin{array}{l} \left|\phi_{j=\frac{3}{2},m}\right\rangle \to 4 \\ \left|\phi_{j=\frac{1}{2},m}\right\rangle \to 2 \end{array}\right\}$ 6 states

expressed in terms of $(2J_1+1)(2J_2+1) = 3\cdot 2 = 6$ states $\left|J_1=1, J_1^z\right\rangle \otimes \left|J_2=\frac{1}{2}, J_2^z\right\rangle$.